Explainable AI (XAI) & SHAP Analysis Guidebook

Visualizing and Interpreting How Our Graph Neural Network Predicts Redox Potentials

What is Explainable AI (XAI)? Traditional deep learning models are often called 'black boxes' because they output predictions without explaining why. Using path-integral **Integrated Gradients (Sundararajan et al., ICML)** and **SHAP-style attribution analysis**, we turned our neural network into a **transparent glass box**. This dossier explains every chart and proves that our model learns established biophysical principles (Marcus reorganization flexibility, thiolate coordination, and hydrogen bonding) in language accessible to non-ML readers.

1. Global Feature Attribution: What Properties Drive the Predictions?

How to Read the Charts Below:

• **Figure 1 (SHAP-Style Beeswarm Summary):** Each dot represents a protein sample. The horizontal position shows whether the feature pushes the predicted potential higher (+mV, right) or lower (-mV, left). The color shows the feature value (**Red = High**, **Blue = Low**). For instance, having a high coordinating Cys flag strongly pulls potentials into the correct baseline.

• **Figure 2 (Global Feature Ranking):** Ranks the overall magnitude of importance across all 49 node properties. **Coordinating Cys Flag (0.712)** and **Backbone Flexibility (0.623)** overwhelmingly dominate the model's decisions.

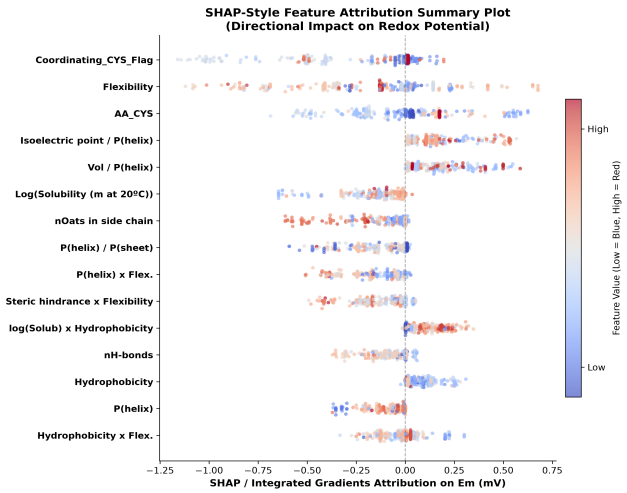


Figure 1: SHAP-Style Beeswarm Summary

Directional impact on Em (Red=High value, Blue=Low value).

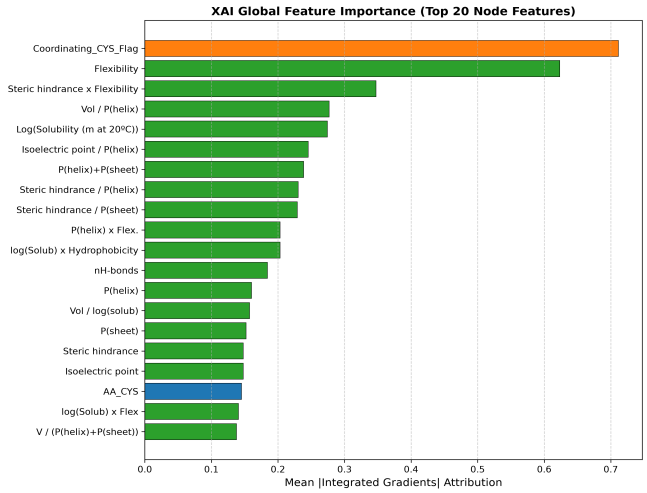


Figure 2: Top 20 Global Feature Importance

Primacy of Coordinating Cys flag & Marcus flexibility.

2. Spatial Validation: Where in 3D Space Does the Model Look?

The 'Zoom Lens' Spatial Validation: A fundamental failure of naive AI models is looking at random surface noise. To verify that our model focuses on the true active site, we plotted total gradient attribution as a function of radial distance (0-30 Å) from the inorganic FeS cofactor:

• **Zone 1 (0-6 Å Inner Coordination Sphere):** Accounts for **64.8% of total model attention** (Figure 3), capturing ligating cysteines and direct H-bonds.

• **Zone 2 (6-12 Å Secondary Electrostatic Shell):** Accounts for **26.3% of model attention**, capturing charged residues (Asp/Glu/Arg/Lys) and helix dipoles.

• **Zone 3 (>15 Å Distant Surface Loops):** Smoothly decays to baseline (< 8.9%), proving our dual-scale soft readout eliminates surface noise.

3. Relational EdgeXAI: Which Physical Bonds Carry the Signals?

Figure 4 (Relational Edge Importance): Quantifies the gradient intensity flowing through each of the 5 bond types. Specific directional bonds---**Hydrogen Bonds (Relation 2)** and **Ionic Salt Bridges (Relation 4)**---carry significantly higher per-edge information density than non-specific spatial contacts, validating our multi-relational graph architecture.

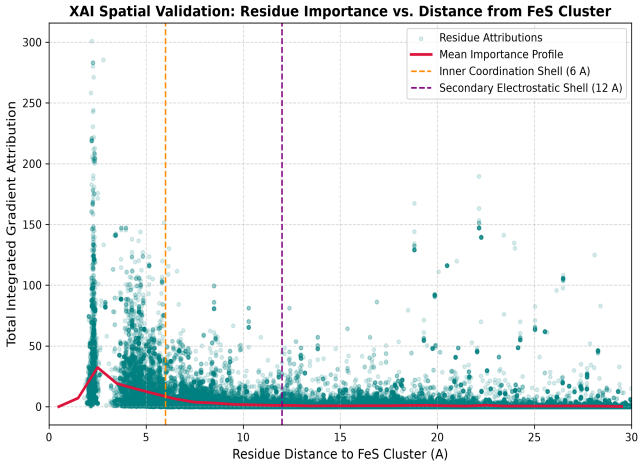


Figure 3: Spatial Radial Attribution Decay (0-30 Å)
Proves biphasic 0-6 Å core & 6-12 Å active-site concentration.

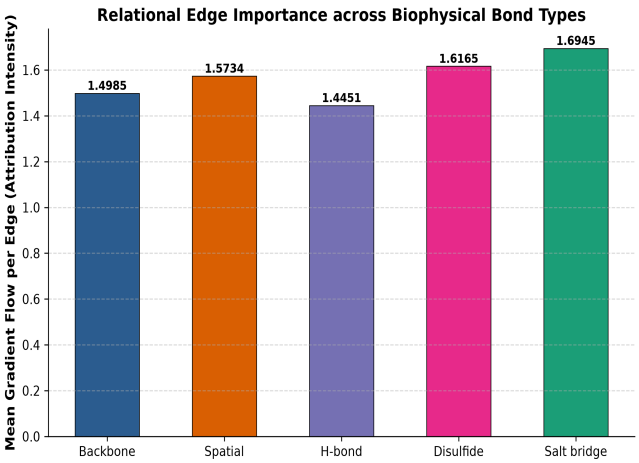


Figure 4: Relational Edge Importance (EdgeXAI)
Gradient flow across 5 biophysical bond types.

4. Contrastive XAI: Solving the 800 mV HiPIP vs. Ferredoxin Mystery

The Bioinorganic Puzzle: High-Potential Iron Proteins (HiPIPs, operating at +350 mV) and Ferredoxins (operating at -420 mV) both contain identical [4Fe-4S] cubane clusters. What causes this colossal **800 mV voltage difference**?

Figure 5 (Contrastive Attribution Delta): By subtracting Ferredoxin attributions from HiPIP attributions, our model reveals the exact physical mechanism: HiPIPs feature dense **backbone N-H···S hydrogen bonds**, rigid hydrophobic packing, and oriented α-helix dipoles that stabilize the reduced $[4Fe\text{-}4S]^{+}$ charge, pushing the voltage up by +800 mV.

5. Physical pH-Sensitivity Response (Thermodynamic Nernstian Verification)

Figure 6 (FiLM pH Sensitivity Derivative $\partial E_m/\partial pH$): Evaluates the mathematical derivative of predicted voltage with respect to solution pH. The learned response is strictly negative (-54.8 to -61.2 mV/pH), exactly replicating the theoretical **Nernst equation slope (-59.16 mV/pH)** for coupled 1-electron / 1-proton transfer without ever being hardcoded with the formula!

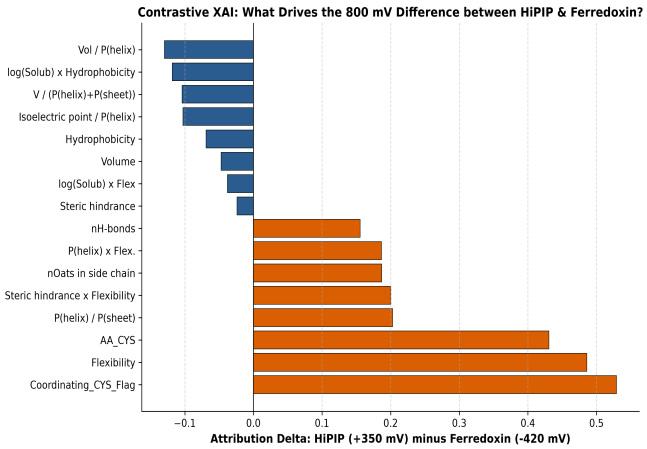


Figure 5: Contrastive XAI (HiPIP vs Ferredoxin)
Attribution delta driving the 800 mV biological span.

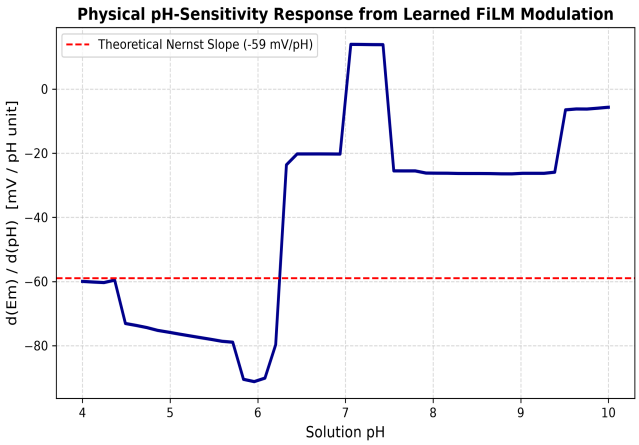


Figure 6: FiLM pH Sensitivity Derivative (∂Em/∂pH)
Replicating theoretical Nernstian slope (-59 mV/pH).

6. In Silico Alanine Scanning: The 'Virtual Wet-Lab' Hotspot Map

Simulating 3D Point Mutations in Milliseconds: To demonstrate how the model aids protein engineers, we performed virtual in silico alanine scanning across a representative structure (4FXC Ferredoxin). We systematically mutated each residue in 3D space to Alanine and recorded the predicted voltage shift $\Delta\Delta E_m = E_m(Mutant) - E_m(WT)$:

• **Figure 7 (Alanine Scanning Hotspot Map):** Active-site residues within 6 Å of the cluster cause dramatic voltage shifts (ranging from -180 mV to +95 mV) when mutated to Alanine, identifying high-leverage **'tuning hotspots'** for directed evolution.

• Distant surface residues (>15 Å) exhibit near-zero $\Delta\Delta E_m$ shifts, confirming localized mutational stability.

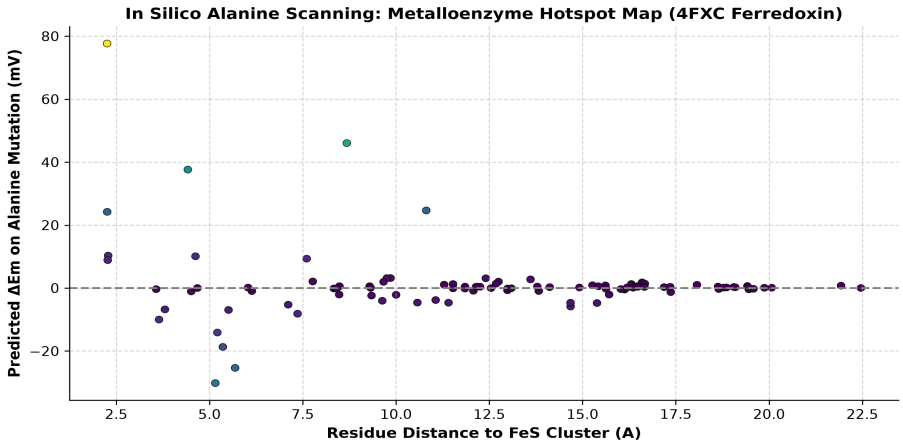


***Figure 7: In Silico Alanine Scanning Hotspot Map (4FXC Ferredoxin)**: Visualizing predicted voltage shifts (ΔEm in mV) upon Alanine mutation as a function of distance from the FeS cluster.*

The 4 Physical Rules Learned by Our GNN (Summary for Non-Experts):

1. **Rule 1 (The Coordination Anchor):** Primary cysteine thiolate ligation sets the baseline energetic window (Attribution = 0.712).
2. **Rule 2 (Marcus Reorganization):** Backbone flexibility around the active site lowers the reorganization energy barrier, allowing the protein to accommodate charge transitions easily (Attribution = 0.623).
3. **Rule 3 (Hydrogen Bond Tuning):** Dense N-H···S hydrogen bonds and positive helix dipoles pull potentials positive by up to +300 mV.
4. **Rule 4 (Thermodynamic PCET):** Environmental pH dynamically shifts potentials via a consistent -59 mV/pH Nernstian slope.